Metabolic Cholestasis, Part Two: Citrin Deficiency and the Inborn Errors of Bile Acid Synthesis.

Picture two infants on the same liver service. The first is a four-week-old girl whose parents are from southern China. Her newborn screen came back flagged for a high blood galactose, so she was tested for classic galactosemia, and the red cell enzyme was normal. Now she is jaundiced, her stools are pale, her albumin is low, her clotting time is prolonged, and her alpha-fetoprotein, the fetal liver protein usually shortened to AFP, is enormous. The second is a two-month-old boy with conjugated jaundice. His gamma-glutamyl transferase, the canalicular enzyme usually shortened to GGT, is only thirty units per liter, which is normal. He does not scratch. His total serum bile acids are low. The team is about to start ursodeoxycholic acid, the bile acid medicine usually shortened to UDCA, because that is what one does for a cholestatic infant.

Both children have inborn errors of metabolism that a routine liver panel can miss. The first has citrin deficiency, which in infancy is called neonatal intrahepatic cholestasis caused by citrin deficiency, shortened to NICCD. The second has a defect in the synthesis of bile acids. The clinical question of this chapter is how to recognize these two groups of disorders inside the ordinary cholestatic infant work-up, and what to do once you suspect them. The main idea is this. Each disorder has a laboratory signature that GGT and bilirubin alone will not reveal. Citrin deficiency looks like neonatal hepatitis with a metabolic tail: high GGT, high bile acids, very high AFP, galactose and the amino acid citrulline raised in the blood. Bile acid synthesis defects look like a low-GGT cholestasis in a child who does not itch and whose measured bile acids are low. And in both, the treatment is something you feed or prescribe: a lactose-free formula rich in medium-chain triglycerides for the first, and oral cholic acid for the second.

We will move through the topic in order. First the mechanism of citrin deficiency, then how common it is, what the infant looks like, how to treat it, what happens later in life, and the sugar trap that makes it dangerous in adulthood. Then the mechanism of bile acid synthesis defects, how to separate them from the other causes of low-GGT cholestasis, how to make the diagnosis, and the long-term results of cholic acid. We finish with a side-by-side comparison, the guidelines, the traps, and what to do differently in clinic.

A note on sources. For the cholestatic infant in general we use the 2017 joint guideline on the evaluation of cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For citrin deficiency we use the Japanese cohort of seventy-five infants reported by Ohura and colleagues in 2007, the Japanese national cohort of two hundred and twenty-two patients reported by Kido and colleagues in 2022, the population

screening study by Lu and colleagues in 2005, and reviews by Hayasaka and Numakura in 2018 and by Makris and colleagues in 2024. For bile acid synthesis defects we use the 2007 review by Heubi, Setchell and Bove from the Cincinnati group, the French long-term cholic acid cohort reported by Gonzales and colleagues in 2018, the Chinese cohort reported by Zhao and colleagues in 2021, the cholic acid phase three trial reported by Heubi and colleagues in 2017, and the regulatory documents for the cholic acid products in the United States and Europe.

Part one. Citrin: one carrier, three consequences.

Citrin is a transport protein in the inner membrane of liver mitochondria. Its job is to move aspartate out of the mitochondrion into the cytosol, in exchange for glutamate moving in. That sounds like a small piece of housekeeping, but it is part of something larger called the malate-aspartate shuttle. The shuttle is how the liver cell moves the reducing power of cytosolic NADH, the reduced form of nicotinamide adenine dinucleotide, into the mitochondrion to be burned. The gene for citrin is called SLC25A13. When both copies are broken, three things follow, and it helps to take them one at a time.

The first consequence is that NADH is trapped in the cytosol. Because the shuttle cannot run, the cytosol becomes over-reduced: too much NADH, too little of its oxidized partner NAD. Glycolysis, which needs NAD to keep turning, slows down. The second consequence concerns the urea cycle. The enzyme argininosuccinate synthetase joins citrulline to aspartate. With less aspartate available in the cytosol, and with the enzyme protein itself reduced, citrulline piles up in the blood and ammonia can rise. The third consequence explains much of the clinical danger. Galactose, fructose and glycerol are all metabolized in the cytosol in steps that generate more NADH. In a cell that is already drowning in NADH, these sugars cannot be processed properly. The reviews describe fructose metabolism stalling at the step of glycerol-3-phosphate dehydrogenase, draining the cell of ATP and phosphate. In the newborn, galactose from milk is handled poorly, which is why the newborn screen can flag galactosemia even though the classic galactosemia enzyme is normal.

There is also a way out. Medium-chain triglycerides, usually shortened to MCT, are fats that enter the mitochondrion without needing a carrier and are oxidized to acetyl-coenzyme A. This raises ATP, restores the ratio of NAD to NADH, and lowers ammonia. So the same biochemistry that makes sugar harmful makes medium-chain fat helpful.

Patients seem to know this instinctively. Older children and adults with citrin deficiency typically eat about fifteen to twenty-five percent of their calories as protein, forty to fifty percent as fat and only thirty to forty percent as carbohydrate, compared with about ten to fifteen percent protein, twenty-five to thirty-five percent fat and fifty to sixty percent carbohydrate in healthy Japanese controls. Their aversion to sweets is not fussiness. It is self-protection.

Citrin deficiency has three faces across life. In infancy it is NICCD. In childhood some patients have failure to thrive and abnormal blood lipids, a stage called failure to thrive and dyslipidemia caused by citrin deficiency. In adults a minority develop adult-onset type two citrullinemia, shortened to CTLN2, a form of hyperammonemic encephalopathy that is potentially fatal.

Part two. How common, and what the infant looks like.

Citrin deficiency is a founder disease of East Asia. In the 2005 screening study by Lu and colleagues, which looked for twelve known Japanese variants, the carrier rate was about one in forty-eight people in China south of the Yangtze River, about one in sixty-five Chinese overall, about one in sixty-nine Japanese, about one in one hundred and twelve Koreans, and only about one in nine hundred and forty in northern China. The authors estimated more than eighty thousand affected homozygotes across East Asia. A later Japanese estimate put the carrier rate at about one in forty-two and the number of affected people at about one in seven thousand one hundred. Two variants were found in every population tested, which means a small targeted panel catches many cases, but the disease has been reported worldwide, so ancestry should raise suspicion without gating testing.

What does the infant look like? In the Japanese national cohort reported by Kido and colleagues, one hundred and ninety-two patients had NICCD. Cholestasis was present in seventy-nine percent. Transaminases above one hundred units per liter were present in seventy-one percent. Low blood protein was present in thirty-nine percent, a prolonged prothrombin time in thirty-four percent, fatty liver in thirty-three percent, poor weight gain in thirty-two percent, low blood sugar in thirty percent and high ammonia in eleven percent. Newborn screening found about a quarter of the cases. In the earlier cohort of seventy-five infants described by Ohura and colleagues, thirty were referred before one month of age because the newborn screen was positive for galactose, methionine or phenylalanine, and forty-five presented with jaundice or pale stools that looked like neonatal hepatitis or even biliary atresia. Plasma citrulline and methionine were significantly raised, and blood galactose was high.

The laboratory signature deserves its own paragraph, because it contradicts a common assumption. A Chinese single-centre study published by Li and colleagues in 2025 compared fifty-five infants with NICCD to twenty-seven infants with idiopathic neonatal cholestasis. The median GGT was about one hundred and eighty-nine units per liter in NICCD against ninety-two in the comparison group. The mean total serum bile acid concentration was two hundred and forty-seven micromoles per liter against one hundred and twenty-seven. And the median AFP was about sixty thousand five hundred nanograms per milliliter against about four thousand two hundred. So NICCD is a high-GGT, high-bile-acid cholestasis with a strikingly high AFP. Yet the 2017 joint guideline, in its table of causes, lists citrin deficiency among the

conditions with normal or only slightly elevated liver enzymes. The lesson is not to be reassured by a modest picture, and not to be put off by a picture that is not modest.

What should you send? Plasma amino acids, looking for raised citrulline, methionine, threonine and tyrosine; blood galactose; AFP; ammonia; glucose; and sequencing of SLC25A13, either on its own or as part of a cholestasis gene panel that includes it.

Part three. Treatment, outcome and the sugar trap.

The treatment of NICCD is dietary. Lactose is removed while galactose is high, using a lactose-free formula. The formula is enriched with MCT, and in the national cohort ninety percent of children aged up to three received an MCT formula. Fat-soluble vitamins A, D, E and K are supplemented, and the coagulopathy is corrected with vitamin K. Then you watch it resolve. In the Ohura cohort, symptoms resolved by twelve months of age in all but two of the seventy-five infants. Those two, about two point seven percent, had progressive liver failure and received a transplant before their first birthday. In the larger national cohort only one of one hundred and ninety-two infants with NICCD, about half of one percent, needed a transplant, there were no deaths from NICCD, and every patient under twenty years of age was alive.

The longer story is about protecting the patient from CTLN2. Fewer than ten percent of people with citrin deficiency develop it, typically between twenty and seventy years of age, with a male to female ratio of about two point four to one. More than ninety percent of them have a body mass index below twenty. Known triggers are surgery, fasting, alcohol and certain medicines. Today a low-carbohydrate diet with MCT has replaced routine liver transplantation as the first approach.

This is where the sugar trap appears. In most urea cycle disorders, the reflex for high ammonia is to stop protein and give a generous glucose infusion. In citrin deficiency that reflex can be harmful. High-sugar infusions add to the cytosolic NADH that the cell cannot clear, and ammonia can climb instead of falling. The reviews go further about glycerol and fructose-containing infusions, which are used in some countries for cerebral edema: Hayasaka and Numakura describe them as fatal and contraindicated in CTLN2, and the 2024 review by Makris and colleagues warns that high-sugar or glycerol infusions may be life-threatening. So in clinic, let the child choose protein and fat and do not force sweets on a child who refuses them. Give the family a written emergency letter naming the diagnosis and the infusions to avoid, and flag the diagnosis in the electronic record, because the crisis may come decades later in a hospital that has never heard of citrin.

Part four. Bile acid synthesis: no primary bile acids, toxic precursors instead.

Now the second infant. The liver makes its two primary bile acids, cholic acid and chenodeoxycholic acid, from cholesterol through a sequence of enzymes. The first step of the

classic pathway adds a hydroxyl group at position seven. Two steps later come the enzymes that matter most in children. The first is three beta hydroxy delta five C27 steroid dehydrogenase, usually called three beta HSD, encoded by the gene HSD3B7. Its deficiency is the commonest bile acid synthesis defect. The second is delta four three oxosteroid five beta reductase, encoded by AKR1D1. Its deficiency is rarer and often more severe.

When either enzyme fails, two harms follow. First, there is no cholic acid to drive bile flow, so bile formation fails and cholestasis results. Second, the precursors that pile up above the block are diverted into atypical bile acids that injure hepatocytes. A third piece completes the loop. Normal bile acids switch off their own synthesis by feedback. Without them there is no brake, so the liver keeps making more of the toxic precursors. Giving cholic acid restores the brake. That is why cholic acid works so well: it does not merely replace what is missing, it turns off production of what is harmful.

The mechanism also explains the two clinical clues. GGT stays normal because the atypical bile acids are poor detergents and do not strip GGT from the canalicular membrane the way high concentrations of normal bile acids do. And the measured serum bile acids are low or normal because the routine assay detects normal bile acids, not the atypical ones. Pruritus, which tracks retained normal bile acids, is usually absent. So the child has conjugated jaundice, normal GGT, low measured bile acids and no itch. Fat-soluble vitamin deficiency is common, and some children present late with rickets or chronic liver disease rather than neonatal jaundice. The delta four reductase defect is an exception to the low GGT rule: GGT can be elevated, and the presentation is often earlier and harsher, with coagulopathy and rapid progression to cirrhosis.

Part five. Separating the low-GGT cholestases.

The useful habit is to read GGT and serum bile acids together. In progressive familial intrahepatic cholestasis types one and two, caused by variants in ATP8B1 and ABCB11, and in tight junction protein two deficiency, GGT is normal or low, but serum bile acids are high and itching is severe. In three beta HSD deficiency, GGT is normal, serum bile acids are low or normal and itching is usually absent. In delta four reductase deficiency, GGT is normal or high, bile acids are low or normal and itching is usually absent. Citrin deficiency, for contrast, has high GGT and high bile acids. So the pair that matters is this: low GGT with high bile acids and itch points to the familial intrahepatic cholestases; low GGT with low bile acids and no itch points to a synthesis defect.

One caution. In the Chinese HSD3B7 cohort reported by Zhao and colleagues, serum total bile acids in infants referred before one year ranged from zero point two to about eighty-five micromoles per liter, and a few were above thirty. So low is a tendency, not a rule, and a moderately raised value does not exclude the diagnosis.

Part six. Making the diagnosis.

Bile acid synthesis defects are estimated to cause about one to two percent of chronic liver disease in infants, children and adolescents. The Cincinnati reference laboratory found one hundred and twenty-eight defects among about seven thousand urine samples over twenty years. The test is urine bile acid analysis by mass spectrometry, either fast atom bombardment mass spectrometry or liquid chromatography with tandem mass spectrometry, which identifies the atypical species directly. The 2017 joint guideline places this test in its second tier and advises considering it as a screening tool before UDCA is started. That timing matters. UDCA and its metabolites flood the urine profile and can hide the diagnosis, so if UDCA has already been started it should be stopped for about seven to ten days before the sample is collected.

There is a second trap with delta four reductase deficiency. In the primary defect, delta four three oxo bile acids make up more than seventy-five percent of urinary bile acids. But the same species appear secondarily in severe liver disease of any cause and in the first weeks of normal life. So a single urine showing delta four acids in a sick infant is not a diagnosis. Repeat the urine after recovery and confirm with sequencing of AKR1D1. Gene panels for cholestasis now include HSD3B7, AKR1D1 and the rarer genes, and they complement the urine test rather than replace it.

Part seven. Treatment with cholic acid.

The longest follow-up comes from France. Gonzales and colleagues reported fifteen patients, thirteen with three beta HSD deficiency and two with delta four reductase deficiency, treated with oral cholic acid for a median of twenty-one point four years and followed to a median age of about twenty-four. All fifteen were alive with their native liver. Liver tests, AFP, vitamin levels and growth were normal at last follow-up. Cirrhosis regressed in three patients. Ten pregnancies were uneventful. Atypical urinary bile acids fell about three hundredfold in three beta HSD deficiency and about thirty-seven-fold in delta four deficiency. Across more than three hundred patient-years there was no serious adverse event. The mean maintenance dose was about six to seven milligrams per kilogram per day, with a cap of five hundred milligrams per day. Overdose had a recognizable signature in the earlier report from the same group: diarrhea, pruritus, and a rise in both GGT and ALT, the alanine aminotransferase, which reversed when the dose was reduced.

The licensing data tell a more sobering story because they include sicker and later patients. The United States cholic acid product, Cholbam, was approved by the Food and Drug Administration in 2015 for bile acid synthesis disorders due to single-enzyme defects and as an adjunct for peroxisomal disorders including the Zellweger spectrum, at ten to fifteen milligrams per kilogram per day. In its label, sixty-four percent of patients with single-enzyme defects responded, twenty-eight of forty-four. Among them, fifty-nine percent of those with three beta HSD deficiency responded, twenty-two of thirty-seven, and three of four with

AKR1D1 deficiency. Only forty-six percent of patients with peroxisomal disorders responded, eleven of twenty-four. Survival beyond three years was sixty-seven percent for single-enzyme defects and forty-two percent for peroxisomal disorders. The phase three program behind it ran for eighteen years and enrolled eighty-five patients; urine bile acid scores and the transaminases improved significantly and there were no drug-related serious adverse events. The label asks for liver tests and the international normalized ratio monthly for three months, every three months for the rest of the first year, every six months for three years and then yearly, and says to stop if liver function has not improved by three months.

In Europe, the European Medicines Agency authorized Orphacol in 2013 for three beta HSD and delta four reductase deficiency from one month of age, at five to fifteen milligrams per kilogram per day. A second product, Kolbam, was authorized in 2014 for other defects and withdrawn for commercial reasons in 2020.

Where cholic acid is not available, chenodeoxycholic acid is used. In the Chinese HSD3B7 cohort, thirty-three children were treated with chenodeoxycholic acid at three to ten milligrams per kilogram per day. Seventy-three percent normalized, six percent improved, fifteen percent needed a liver transplant and six percent died. The median age at onset in that cohort was ten days, thirty-two of thirty-nine presented with neonatal cholestasis, GGT in infants never exceeded seventy units per liter, and more than a quarter had kidney lesions before treatment. The transplants and deaths were driven by late referral and established cirrhosis, which is the strongest argument for sending the urine early.

UDCA is not the answer. It can lower the transaminases, but it does not suppress synthesis of the atypical bile acids, and in the Cincinnati experience it worsened liver tests in one patient and blunted the effect of cholic acid when the two were combined. A rarer defect, of oxysterol seven alpha hydroxylase encoded by CYP7B1, presents with neonatal liver failure. The first reported infant did not respond to cholic acid and needed a transplant, while a later infant responded to chenodeoxycholic acid at six milligrams per kilogram per day.

Part eight. Side by side, and what the guidelines say.

Putting the three main disorders next to each other: NICCD presents in the first weeks, often through the newborn screen, with high GGT, high bile acids, variable itch, and a signature of raised citrulline, galactose and a very high AFP. The key tests are plasma amino acids and SLC25A13, the treatment is lactose-free MCT formula, and the long-term issue is the risk of CTLN2. Three beta HSD deficiency presents at a median of about ten days, sometimes much later, with normal GGT, low or normal bile acids, no itch, and sometimes rickets or vitamin deficiency. The key tests are urine bile acids and HSD3B7, the treatment is cholic acid, and the outlook on treatment is a normal life. Delta four reductase deficiency presents early and severely, with normal or high GGT, low or normal bile acids, coagulopathy and rapid fibrosis. The key tests are repeated urine bile acids and AKR1D1, the treatment is cholic acid, and the

outlook is good if treatment starts early.

On the guidelines, North America and Europe share one document for the cholestatic infant, the 2017 joint guideline, so they are aligned on screening with urine bile acids before UDCA and on the point that serum bile acids are usually low in synthesis defects. They differ on drugs, because the regulators differ: the United States label covers single-enzyme defects and peroxisomal disorders at ten to fifteen milligrams per kilogram per day, while the European product covers the two main defects from one month of age at five to fifteen milligrams per kilogram per day and does not cover peroxisomal disorders. There is one material gap: the joint guideline describes citrin deficiency as having normal or slightly raised enzymes, while the cohort data show high GGT and transaminases above one hundred in most infants. For the management of citrin deficiency there is no society guideline on either side of the Atlantic, only expert reviews, and neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has specific guidance on these disorders.

Part nine. Traps and pearls.

Four traps account for most late diagnoses. The first is thinking that a normal GGT means mild disease. Low-GGT cholestasis is often severe, and the answer is to measure serum bile acids. The second is starting UDCA first, which makes the urine profile uninterpretable; stop it for seven to ten days and repeat, or better, send the urine before the first dose. The third is diagnosing delta four reductase deficiency on a single urine from a sick infant, when the pattern may be secondary; repeat it and sequence the gene. The fourth is calling an East Asian infant with galactosemia on the screen and a normal galactosemia enzyme just neonatal hepatitis; look at citrulline and AFP and include SLC25A13 in the panel.

The numbers worth carrying into clinic are these. Citrin deficiency carriers number about one in forty-eight in southern China, one in sixty-five Chinese overall and one in sixty-nine Japanese. In NICCD, cholestasis is present in seventy-nine percent, low protein in thirty-nine percent and low blood sugar in thirty percent, with galactose and citrulline raised. NICCD has high GGT, high bile acids and a very high AFP. On lactose-free MCT formula almost all infants recover by twelve months, and transplantation is needed in about half a percent to under three percent. Glycerol, fructose and high-sugar infusions should be avoided. Bile acid synthesis defects cause one to two percent of pediatric liver disease and present with normal GGT, low bile acids and no itch. Urine bile acids should be sent before UDCA. And on cholic acid, fifteen of fifteen French patients kept their native liver after a median of twenty-one years.

What to do differently in clinic.

When you meet a cholestatic infant, look at GGT and serum bile acids together, and ask about itch. If GGT is normal, bile acids are low and the child does not scratch, send urine bile acids

by mass spectrometry before anyone writes a prescription for UDCA, and start cholic acid once the diagnosis is confirmed, watching for the combined rise of GGT and ALT that signals too much. If the newborn screen flagged galactose but the galactosemia enzyme is normal, or if the infant has East Asian ancestry, low protein and a remarkable AFP, send plasma amino acids and SLC25A13, switch to a lactose-free MCT formula, and expect recovery within the year. Then think ahead twenty years: write the diagnosis and the infusions to avoid into an emergency letter, so that the next team, faced with a confused adult after surgery, does not reach for the sugar.

End of chapter.